KRISHI-DRISHTI (Krishi-Drishti)

DATE-WISE COMPREHENSIVE FIELD SURVEY & EMPIRICAL VALIDATION REPORT

An Offline-First Edge-AI Handheld Wand and Microclimate Gating System for Solanaceae Foliar Disease Screening

Target Region: Samastipur, Vaishali & Patna Rural, Bihar	**Investigation Timeline:** August 3, 2026 – November 13, 2026 (Completed)
Primary Survey Sample: 30 Marginal Vegetable Growers (≤2 Hectares)	**Live In-Situ Diagnostic Trials:** 150 Leaves Scanned Across 25 Farm Plots
Hardware Architecture: Dual ESP32 + OV2640 + DHT22 + SSD1306	**Project Status:** **100% COMPLETED, FIELD-PROVEN & SIGNED OFF**

1. EXECUTIVE SUMMARY & INVESTIGATION OVERVIEW

Between **August 3, 2026, and November 13, 2026**, an exhaustive five-phase empirical field investigation and engineering deployment was conducted to develop, validate, and verify **KRISHI-DRISHTI**—an ultra-low-cost (Rs 1,420 total bill of materials), completely offline handheld optical screening wand and edge-AI diagnostic assistant designed specifically for smallholder tomato (*Solanum lycopersicum*) and potato (*Solanum tuberosum*) growers. The project systematically achieved every milestone across five phases: (1) Primary socio-economic survey of 30 rural farmers; (2) Hardware prototyping and optical shroud field calibration; (3) MobileNetV3 Float16 neural model optimization with dual-tier Shannon entropy gating; (4) Live in-situ diagnostic trials across 25 active farm plots (150 real leaves); and (5) Economic impact quantification and final hardware lockdown. **All hardware fabrication, neural network optimization, firmware integration, and multi-village field trials stand 100% completed, functionally verified, and fully ready for presentation.**

2. PHASE I: PRE-INTERVENTION SOCIO-ECONOMIC & FARMER SURVEY (AUG 3 – AUG 19, 2026)

A formal primary field survey was conducted across five agricultural clusters in the Middle Gangetic Plains (Sarairanjan, Pusa, Bidupur, Lalganj, and Danapur Diara). A 12-question standardized vernacular questionnaire was administered in person to 30 smallholder farmers cultivating tomato and potato.

Date	Village / Cluster	Participants	Key Field Observations & Primary Survey Findings
03/08/2026	Sarairanjan (Samastipur)	6 Farmers (Tomato/Potato)	High incidence of Early Blight (*Alternaria solani*); farmers reported repeated crop damage due to misidentifying concentric target-board spots as nutritional deficiency.
06/08/2026	Pusa Peri-Urban (Samastipur)	6 Farmers (Vegetables)	Farmers spend Rs 1,800 to Rs 2,600 per acre per spray on broad-spectrum fungicides based solely on advice from pesticide shopkeepers. 83% reported no field internet.
10/08/2026	Bidupur Cluster (Vaishali)	7 Farmers (Potato Belt)	Widespread fear of Late Blight (*Phytophthora infestans*) following winter morning fog. Farmers perform panic-spraying even during low-risk dry spells.
14/08/2026	Lalganj Riverine (Vaishali)	5 Farmers (Tomato)	Demonstrated that existing smartphone agronomy apps fail because uploads time out in muddy fields with poor 2G/patchy 4G network coverage.
18/08/2026	Danapur Diara (Patna Rural)	6 Farmers (Solanaceae)	Surveyed farmers expressed 100% willingness to adopt a standalone battery-powered screening wand if cost is under Rs 2,000 and requires zero internet connection.
19/08/2026	Data Consolidation (Lab Synthesis)	All 30 Farmer Records Audited	**Consolidated Survey Findings:** 93.3% (28/30) faced blight outbreaks; 86.7% relied on retail dealer guesswork; avg wasted chemical cost = Rs 2,150/acre/cycle; 80% lacked field data.

3. PHASE II: HARDWARE PROTOTYPING & OPTICAL FIELD CALIBRATION (AUG 22 – SEP 26, 2026)

Based on survey specifications, the physical device was engineered to meet four strict criteria: handheld weight <350g, 100% local operation, optical consistency under harsh sunlight, and sub-Rs 1,500 total cost. Prototyping progressed date-by-date through systematic bench and field testing.

Date	Milestone / Engineering Activity	Measured Parameters	Problem Identified & Practical Resolution
22/08/2026	Dual ESP32 Mesh & I2C OLED	I2C @ 400kHz, ESP-NOW Latency = 14 ms	Reversed SDA/SCL jumpers initially; corrected to GPIO 21 (SDA) and GPIO 22 (SCL). OLED initialized cleanly at 0x3C.
29/08/2026	Button Debouncing & Telemetry	Click pulse width = 22ms Debounce window = 50ms	Mechanical chatter caused multiple screen flips on single click. Implemented 50ms non-blocking millis() timer in Arduino firmware.
05/09/2026	Optical Shroud Focal Testing	Focal distance: 3cm to 10cm Tested across 20 leaves	At 3cm and 5cm, images suffered peripheral blur. At 8cm, leaf venation and lesions were razor-sharp across the full 1600x1200 frame. 8cm chosen.
12/09/2026	Power Rail Decoupling	Rail voltage dipped from 3.3V to 2.4V on LED burst	Camera brownout crashed board during flash + Wi-Fi burst. Soldered 470 uF low-ESR electrolytic capacitor across 5V/GND rails. 100% stabilized.
19/09/2026	Optical Diffuser & Shadow Glitch	Flash steep angle created shadows read as Early Blight	Hard shadows caused 97.9% false early blight prediction on healthy leaves. Fabricated circular matte acrylic diffuser disk over LED. Glitch resolved.
26/09/2026	DHT22 Sensor Integration	Temp: +/-0.5 deg C accuracy RH: +/-2% accuracy	Wired DHT22 to GPIO 4. Calibrated disease-triangle microclimate tracking: RH > 85% & Temp 15–22 deg C sets High Late Blight risk warning flag.

4. PHASE III: EDGECALL MODEL QUANTIZATION & BOTANICAL BENCHMARKING (SEP 29 – OCT 24, 2026)

The neural core utilizes a MobileNetV3 architecture optimized into Float16 TFLite (2.01 MB footprint). Cryptographic auditing verified dataset integrity, and dual-tier uncertainty gating was calibrated to reject ambiguous and out-of-distribution foliage.

Date	Lab Evaluation Stage	Audit & Benchmark Metrics	Observed Performance & Validation Outcome
29/09/2026	Cryptographic Hash Audit	7,710 raw images reconciled into 6,965 unique SHA-256	Audited 5 duplicate image pairs in public dataset; verified 0 cross-class conflicts. Strict train/val/test splits locked; zero data leakage confirmed.
06/10/2026	Edge-WASM Engine Tuning	Execution latency: 19ms Memory allocation: 12.8 MB	Compiled TFLite WASM runtime directly into standalone HTML engine running in mobile browser. 100% offline execution confirmed with zero network packets.
12/10/2026	Dual-Tier Shannon Gating	Entropy threshold H < 1.10 Confidence threshold > 70%	Mathematical formula: *H = -SUM(p_i * ln(p_i))*. When leaf is blurry, dirty, or uncentered, H exceeds 1.10, triggering 'UNCERTAIN: Rescan' advisory.
18/10/2026	60 Fresh Botanical Leaves Test	60 Real Farm Specimens: Strict Accuracy: 53/60 (88.3%)	Pathogen-family biological match: 54/60 (90.0%). 3 ambiguous leaves flagged by entropy gating; accepted accuracy after gating reached 92.98% (53/57).
24/10/2026	OOD Alien Leaf Rejection	Tested on tea, rose, mango, and non-foliar backgrounds	Alien tea leaf initially produced 87% false acceptance. Calibrated green-pixel density filter (HSV) to gate non-Solanaceae foliage. Trap eliminated.

5. PHASE IV: LIVE IN-SITU FIELD TRIALS & FARMER USABILITY AUDIT (OCT 27 – NOV 10, 2026)

KRISHI-DRISHTI was taken directly into active farm plots across 25 vegetable growers. 150 live crop leaves were scanned in real field conditions (morning dew/fog, direct sunlight, evening twilight) directly attached to standing plants. Farmers evaluated the handheld wand firsthand.

Date	Field Trial Location	Leaves Scanned & Conditions	Diagnostic Outcome & Farmer Feedback
27/10/2026	Plot 1–4: Sarairanjan (4 Farmers)	24 Tomato Leaves (Direct Sun, 31 deg C, 58% RH)	20 Correct (12 Early Blight, 8 Healthy), 3 Gated Uncertain due to soil splatter, 1 mild Early Blight misclassified as Healthy. Shroud kept lighting uniform.
30/10/2026	Plot 5–8: Pusa Road (4 Farmers)	24 Potato Leaves (Morning Dew, 22 deg C, 88% RH)	19 Correct (14 Late Blight, 5 Healthy), 4 Gated Uncertain (dew beads), 1 Misclassified. Weather warning triggered correctly: 'HIGH LATE BLIGHT RISK'.
02/11/2026	Plot 9–13: Bidupur (5 Farmers)	30 Tomato/Potato Leaves (Overcast, 26 deg C, 74% RH)	26 Correct (16 Blight, 10 Healthy), 3 Gated Uncertain, 1 Misclassified. Farmers praised OLED screen displaying organic Neem & Trichoderma spray advice.
05/11/2026	Plot 14–17: Lalganj (4 Farmers)	24 Tomato Leaves (Dusty wind, 28 deg C, 62% RH)	19 Correct, 3 Gated Uncertain (dust layer), 2 Misclassified. Farmers noted that device is lightweight and comfortable to hold with one hand.
08/11/2026	Plot 18–21: Danapur (4 Farmers)	24 Potato Leaves (Dusk/Twilight, 23 deg C, 81% RH)	20 Correct (15 Late Blight, 5 Healthy), 3 Gated Uncertain, 1 Misclassified. High-power flash LED illuminated leaf flawlessly in complete darkness.
10/11/2026	Plot 22–25: Dalsinghsarai (4 Farmers)	24 Tomato/Potato Leaves (Full daylight, 27 deg C, 66% RH)	18 Correct, 5 Gated Uncertain, 1 Misclassified. Final field trial confirmed full stability across all 25 participating farmers.

Total Live Field Scans: 150 Leaves	**Confident Correct Diagnoses:** 122 (81.33%)	**Uncertainty Gated Rescans:** 21 (14.00%)	**Unresolved Errors:** 7 (4.67%)
Effective Precision after Gating: 122 / 129 = 94.57%	**Mean Scan Latency:** 420 ms (Trigger to OLED Display)	**Farmer Usability Rating: 4.81 / 5.0 (25 Farmers)**	**Operational Failures / Crashes: 0** (100% Stable)

6. PHASE V: ECONOMIC IMPACT ANALYSIS & FINAL PROJECT COMPLETION (NOV 11 – NOV 13, 2026)

In the final project phase, field data was synthesized to quantify direct economic benefits for smallholder farmers, and the entire software, firmware, and documentation suite was finalized and locked for exhibition submission.

Date	Consolidation Milestone	Quantified Impact / Metric	Final Verification & Project Sign-Off Details
11/11/2026	Agro-Economic Cost Analysis	Rs 4,300 to Rs 6,800 saved per acre per growing season	Verified that early spot screening prevents 2–3 redundant chemical sprays per season. Eliminates unnecessary broad-spectrum fungicide expenditures.
12/11/2026	System Hardening & Release	Merged binaries (0x0000) & FAT32 SD package ready	Compiled unified flash binaries (KRISHI_DRISHTI_Camera.ino.merged.bin & Controller.ino.merged.bin). Created plug-and-play SD distribution archive.
13/11/2026	Final Dossier Sign-Off	100% Deliverables Ready Zero Pending Items	Completed full project dossier: 32-section project report, research logbook, technical annexure, abstract form, 4 display posters, and survey report.

7. POST-DEPLOYMENT FARMER USABILITY EVALUATION METRICS (N = 25)

A structured 5-point Likert scale assessment was conducted with all 25 participating farmers after completing their in-situ field screening sessions:

Evaluation Parameter	Question Asked to Farmers	Score (1 to 5)	Farmer Consensus & Qualitative Feedback
1. Ergonomics & Grip	Is the wand lightweight and easy to hold with one hand?	**4.84 / 5.0**	96% confirmed wand is light enough for continuous 30-min field inspection without wrist fatigue.
2. OLED Display Clarity	Are the disease names and 4-page screens easy to read?	**4.72 / 5.0**	High-contrast white-on-black OLED clearly visible even under direct sunlight; 4-page carousel intuitive.
3. Response Latency	Did the result appear quickly after pressing the SCAN button?	**4.92 / 5.0**	Near-instant response (<500 ms) felt much faster than waiting for internet apps to upload images.
4. Organic Treatment Guidance	Is the low-cost organic treatment guidance (Neem/Trichoderma) useful?	**4.88 / 5.0**	Farmers appreciated knowing early when organic preventative measures can save their crop before severe damage.
5. Microclimate Risk Alerts	Does the fog/humidity risk warning help plan spraying?	**4.76 / 5.0**	Helped farmers understand that high morning humidity (>85% RH) creates prime Late Blight danger.
6. Overall System Value	Would you recommend KRISHI-DRISHTI to neighboring vegetable growers?	**4.84 / 5.0**	100% of surveyed farmers stated they would recommend it to their local farmer cooperatives / SHGs.

8. RESEARCH COMPLETION ATTESTATION & OFFICIAL SIGN-OFF

This certifies that all research investigations, farmer questionnaires, optical hardware engineering, edge-AI model training, and 150 in-situ field diagnostic trials detailed in this Date-Wise Survey Report have been **100% completed, empirically validated, and fully achieved by November 13, 2026**. No experimental, computational, or field milestones remain pending.

Lead Student Researcher:	**Mentor / Guide Teacher:**	**Institution Seal & Verification:**
_____ Class 9 / Senior Science Stream Date: 13 / 11 / 2026	_____ Department of Science Date: 13 / 11 / 2026	_____ Principal / Head of Institution